WILDLIFE CONSERVATION USING DRONES TO TRACK WILDLIFE INCIDENTS (POACHING, BIRDS KILLED BY POWER LINES)

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Inspiration

WildTrack has a **unique and award-winning conservation model** based on totally non-invasive methods and community-friendly approaches.

The approaches would include

Terrain mapping

High-resolution imaging

Anti-poaching analysis,

Investigation of inaccessible areas

Wildtrack Experts in the field in Namibia.

Applying a suite of geographical information system (GIS) tools can improve monitoring for elephant poaching, according to Penn State researchers. Investigating Tsavo, a region in southwest Kenya . They sought to understand geographic similarities among elephant poaching locations, analyzing the geographic features of 156 known locations occurring in the 8,150-square-mile area in the Tsavo region.

https://www.psu.edu/news/research/story/catch-poacher-gis-drones-can-improve-elephant-conservation/

Wildlife conservation is a crucial step in preserving the ecology. Since the ecology is deteriorating at a fast pace, governments across the world are adopting various methods to rapidly reverse the damage.

UAV's and drones are some of the innovations that make various aspects of wildlife conservation, like monitoring, tracking, patrolling, and extracting data, much easier.

What is GIS?

- GIS software: A Geographic Information System (GIS Software) is designed **to store, retrieve, manage, display, and analyze all types of geographic and spatial data**. GIS software lets you produce maps and other graphic displays of geographic information for analysis and presentation.
- How does GIS work?
  - GIS **connects data to a map, integrating location data (where things are) with all types of descriptive information (what things are like there)**. This provides a foundation for mapping and analysis that is used in science and almost every industry.

What are UAV's?

UAVs are a component of an unmanned aircraft system, which includes adding a ground-based controller and a system of communications with the UAV.

Professionals use artificial intelligence and machine learning technologies obtained from top tech firms to offer custom UAV solutions to study the animal species present in nature. UAV-powered AI solutions help detect species and protect them from targeted areas where poachers are willing to hunt.

Why use UAV's/drones for Wildlife Tracking?

- In wildlife, drones are used **to count, study, and protect animals from harm**. They are far more accurate at counting species and can record vast amounts of data quicker than a person could. Drones can also capture images from a bird's-eye view, giving them the edge over traditional field research.
- Wildlife monitoring has become challenging for environmental activities like tracking the animals and natural resources present in a forest and looking after wildlife's social environments. The advanced features like altimeters, cameras, and GPS tracking present in UAV based solutions have become a driving force for wildlife monitoring. The UAV based solutions have stronger thermal and long-distance focus point cameras for round-the-clock monitoring of animals. Data engineering consulting firms provide integrated accelerators and motion sensors for UAV drones that collect accurate data on vehicle movement and wildlife, such as birds or animals.

Why was this region chosen?

- According to Joseph Bishop, "The idea of using a drone in an area where it's challenging to do conservation work is coming up more and more because, in the case of Africa, where you've got multiple large animals that move long distances day to day, the idea of trying to monitor them and protect them is daunting,". He added on that "Using drones doesn't cost as much as using a helicopter, and it also keeps people out of danger from being attacked by poachers.”
- The research paper centers about performing a geospatial analyses on the physical environment to create a risk map based on how roads, water features, and land cover correlate to poaching incidents. It also models drone flight paths for surveillance of high-risk areas based on aircraft flight characteristics and the horizontal field of view for a selected infrared camera.

Why is this topic relevant to the world today?

- World elephant populations are declining rapidly due to illegal poaching and trade in ivory. This illegal ivory trade is fueled mostly by East Asia's demand for ivory trinkets and statues, which are a symbol of wealth and social status.
- In addition to threatening the existence of elephants, profits from the illegal ivory trade have been linked to funding crime syndicates and terrorist groups in Africa . The African elephant (Loxodonta africana) is being illegally killed at unsustainable rates, which resulted in a 64 % reduction of the Central African elephant population from 2002 to 2012 . While other human activities also negatively affect the elephant population, such as retaliatory killings by farmers suffering damaged crops, and habitat loss from development , poaching for ivory is currently the greatest threat.

Why are drones better than traditional methods of tracking?

Conventional Methods

- Conventional aerial surveillance of wildlife can cost up to US$50,000 dollars a week.
- Due to the cost, only large funded conservation groups can help contribute to preserve wildlife.
- More manual labor/people have to be involved with risk to life.

Drones

- On the other hand, the drone costs as little as US$3,000 per week .
- The lower cost allows for even the smallest of conservation groups to partake in preserving wildlife.
- The use of UAVs reduce the risk of pilots and conservation experts in the field.

Methodology in the Research Paper

- Study area: The research papers analysis is focused on the protected area complex in the southeastern region of Kenya (Fig. 1) referred to as the Tsavo ecosystem. The Tsavo ecosystem is composed of the Tsavo East National Park, Tsavo West National Park, Chyulu Hills National Park, Ngai Ndethya National Reserve, Taita Hills Wildlife Sanctuary, and the Lumo Wildlife Sanctuary. The entire ecosystem contains 21,128 km^2 of mostly dry, flat, savannah vegetation

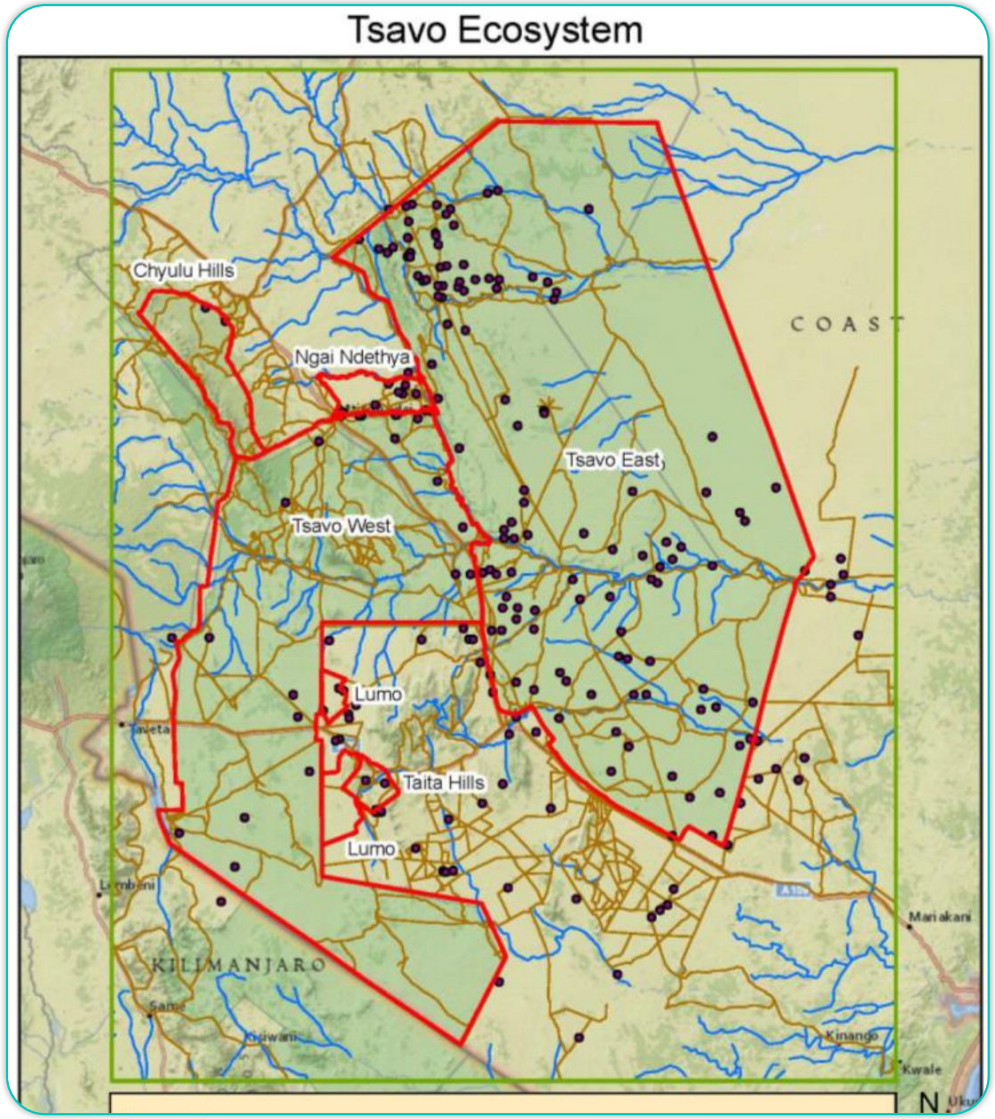


Figure (1)

Data Analysis

- The project uses ESRI's ArcGIS software which is an Artificial Intelligence Geographical Information System (AIGIS), along with the Spatial Analyst Extension to perform the majority of analyses.

- AIGIS tools were used
  - to determine the relationship between poaching incidents and physical environmental characteristics in our study area
  - to model drone surveillance areas
  - to determine the strategic placement of ranger guard posts in response to identified high-risk areas.

- The project also used open-source statistical software, including R-STAT and GeoDa, to perform the point pattern analysis and spatial autocorrelation analysis respectively.

Point pattern analysis on elephant poaching incidents

- Spatial statistical analysis offers tools for identifying the patterns and characteristics of geographic point data.

- This type of analysis can reveal whether point distributions are connected to underlying properties of the physical environment (first-order effects) and whether a point has a positive or negative effect on the presence of additional points (second-order effects).

- Spatial analysis is also used to determine if the point pattern being analyzed is a realization of a deterministic process or a stochastic (random) process. This is necessary as stochastic processes don't permit predictive modeling.To eliminate the possibility that the data resulting from a stochastic process, 100 Monte Carlo simulations were performed.

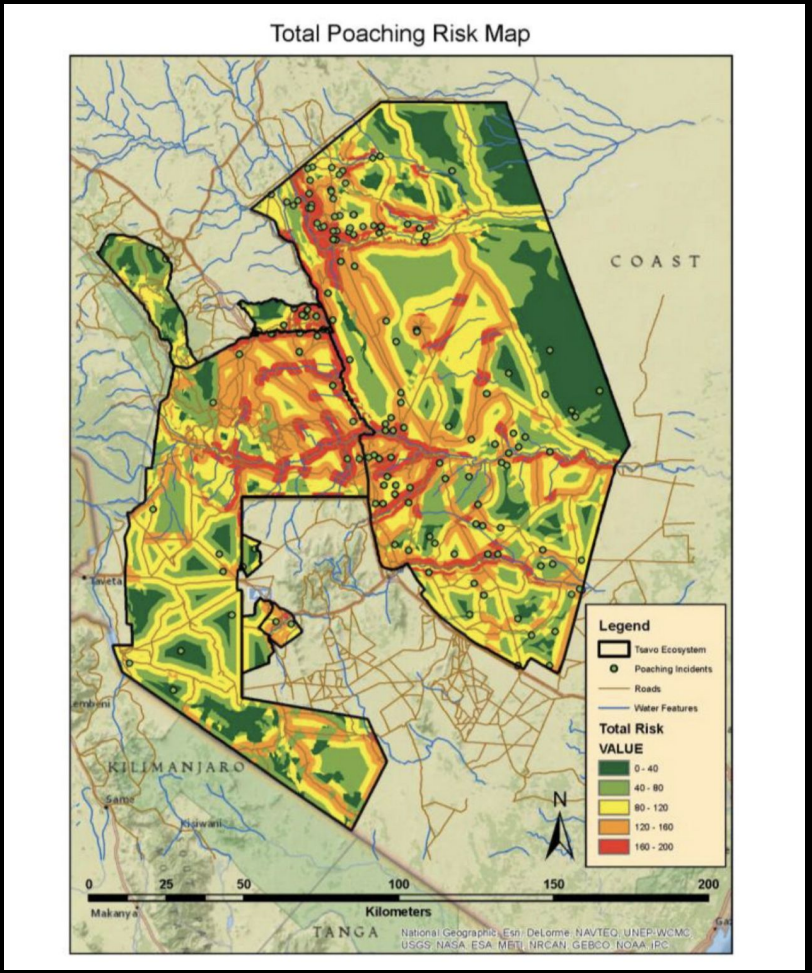
- Each of these analyses was performed on the 156 elephant poaching incident points provided by a Kenyan conservation group.

Identifying poaching high risk areas based on physical environment and incident intensity

- The AI focused on land cover types, proximity to water sources, and proximity to roads in order to find a positive correlation between physical characteristics of the environment in the Tsavo ecosystem and the occurrence of poaching incidents.

- Poaching risk values were assigned to each physical environment feature based on the total number of poaching incidents that occurred within that feature.

- These risk numbers helped determine that poaching incidents are more likely to occur within close proximity to roads due to the easy access to wildlife they provide, and to facilitate escape after poaching has occurred.



Surveillance drone flight paths

- The increasing popularity of drone use has led to the creation of the free open source flight planning software, ArduPilot Mega Planner, which assists users in determining flight mission capabilities.

- This model was fed to the AI program to construct the drone flight paths required to cover a single poaching area that had been determined high risk.

- The inputs of the flight path model included cruising altitude, cruising speed, and flight time at cruising speed, to determine the total distance a drone could fly on one fueling (battery or gas tank).

- The surveillance area covered by the drone was determined by the field of view width from the attached camera used for spotting poachers and the line distance the drone is able to fly.

Fig 3: ArduPilot Mega Planner home screen

Ground crew station locations

- Finally AI assisted in determining the most efficient placement for guard stations to determine their response to poaching threats in high risk areas.
- The program based this on analysis of the time it took to travel over the conservation area terrain given the roads, land cover type, and the presence of water features.
- Roads were assigned the least amount of associated travel cost, while densely forested land covers and river water features were assigned the highest.

Inspiring Features

- Benefit to society
  - World elephant populations are declining rapidly due to illegal poaching and trade in ivory. Global trade for illegal wildlife generates approximately US$10 billion per year, making it the fifth most profitable illicit trade
  - Tracking wildlife and intercepting poachers is more challenging when the majority of the area is covered in dense vegetation, The advancement of UAVs better known as drones is revolutionizing the ability to track and identify poachers.
  - Using drones is quite cost effective and it keeps the people (wildlife rangers) safe from poacher attacks.
  - This technology can be further implemented into different fields for security purposes.

Some numbers…

- 64 % reduction of the Central African elephant population from 2002 to 2012
- In the 19th century, there were between 3 to 5-million elephants
- African elephants have been estimated to routinely travel between 30 km and 60 km in a single day
- The entire ecosystem contains 21,128 km2 of mostly dry, flat, savannah vegetation
- The value of one drone's work is equivalent to 50 rangers
- Conventional aerial surveillance of wildlife can cost up to US$50,000 dollars a week, whereas a drone costs as little as US$3,000 per week
- A recent study in central Africa revealed that prioritizing ranger patrols to focus on the most threatened conservation areas can reduce ranger patrol costs by 63 %
- The horizontal field of view of the drones at an altitude of 91m would be approximately 187 m
  - Total linear distance coverage - 89.3 km

Benefits of using AI in Wildlife Tracking:

1) Animal Tracking

Through foot prints,, droppings, and other animal samples, we are able to track animals and help preserve their habitats

2) Increased biodiversity

AI has given us the ability to trackdown endangered species and return them to their homes

3) Detecting Poachers

With the aid of night vision cameras and thermal radars, AI-drones can detect poachers and swiftly alert rangers accurately of their locations

4) Sound Surveillance

The ability to hear if anything is out of place in a particular location can help early detections of fires, poachers, etc.

Final Thoughts

- Drones display what the camera captures
- AI allows for motion, perception and learning into drones
- Challenges of AI for wildlife
- Understanding the ethical risks involved

Articles/Research Papers Referenced

- Penn State graduate student Jeff Kerby used drones for his ecological research in Greenland and is sharing his expertise to enhance research and conservation efforts worldwide.
- Penn State researchers investigated ways to increase elephant conservation by identifying common geographical characteristics among poaching locations. They then created flight paths for drones that would survey the high-risk poaching areas, which is a technique local conservation agencies can use to optimize their efforts.
- Predicting and Preventing Elephant Poaching Incidents through Statistical Analysis, GIS-Based Risk Analysis, and Aerial Surveillance Flight Path Modeling – Research Paper By Michael Shaffer & Joseph Bishop.

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